

Forecasting Seasonal Changepoint Models Utilizing Changepoint Uncertainties

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Lancs Cpt April 2026

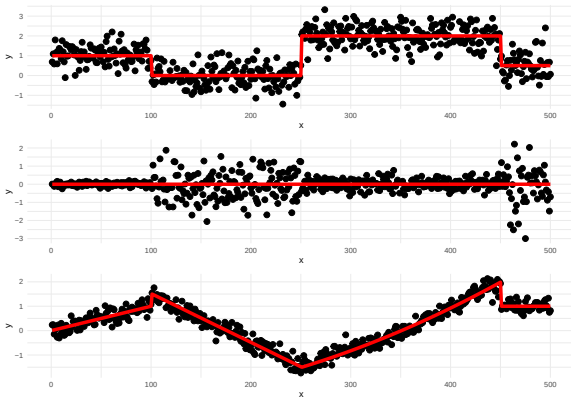
- The challenge
- Blended Forecasting
- Tools required
- Examples

What are changepoints?



For data $y_1, \dots, y_n$, if a changepoint exists at τ , then $y_1, \dots, y_\tau$ differ from $y_{\tau+1}, \dots, y_n$ in some way.

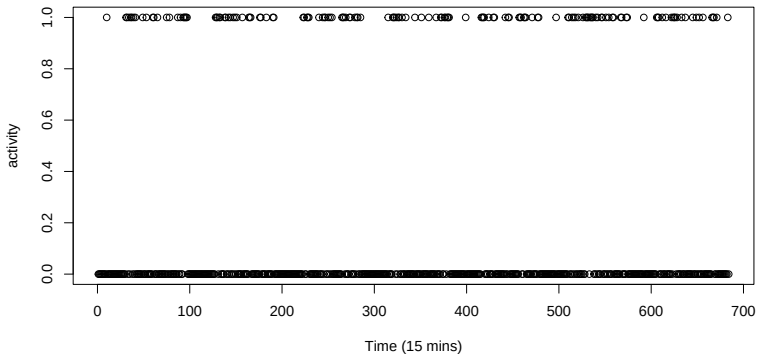
There are many different types of change.



The challenge



To forecast:

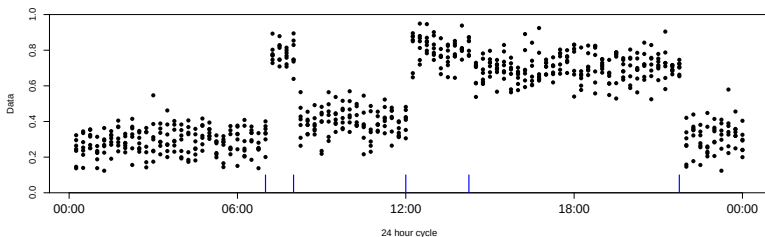
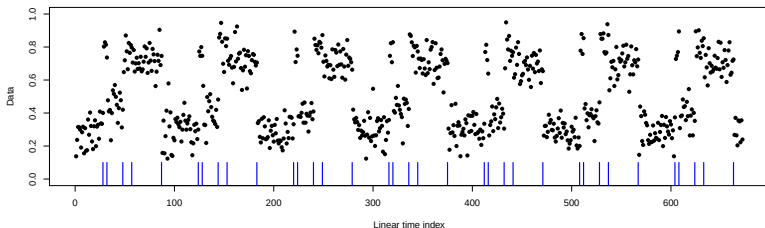




Common choices:

1. Find the last changepoint and forecast from the final segment
2. Find the changepoints, get residuals and forecast the residuals
3. Include the (known) changepoints in a model and forecast

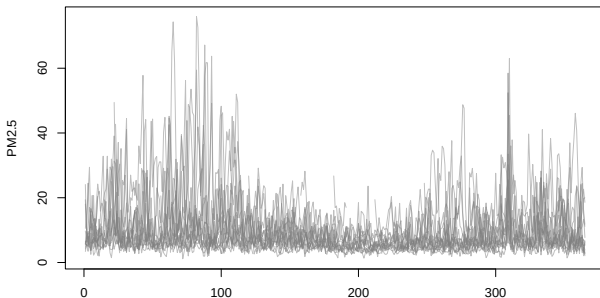
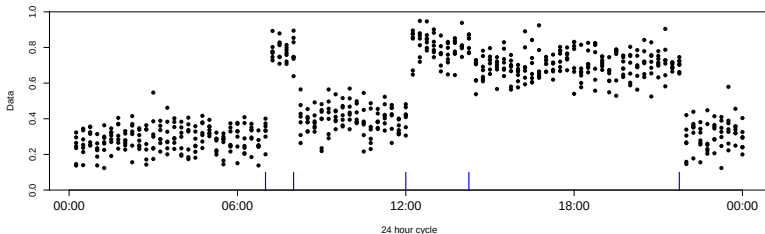
The challenge



The challenge

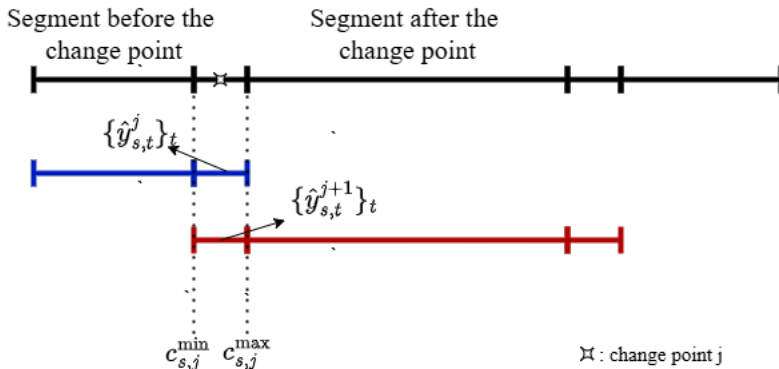
$$\left[\begin{array}{ccccc|c} y_{1,1} & y_{1,2} & \dots & \dots & y_{1,N} & \hat{y}_{1,N+1} \\ y_{2,1} & y_{2,2} & \dots & \dots & y_{2,N} & \hat{y}_{2,N+1} \\ \vdots & \vdots & \ddots & & \vdots & \vdots \\ \vdots & \vdots & & \ddots & \vdots & \vdots \\ y_{p-1,1} & y_{p-1,2} & \dots & \dots & y_{p-1,N} & \hat{y}_{p-1,N+1} \\ y_{p,1} & y_{p,2} & \dots & \dots & y_{p,N} & \hat{y}_{p,N+1} \\ \hline \hat{y}_{p+1,1} & \hat{y}_{p+1,2} & \dots & \dots & \hat{y}_{p+1,N} & \end{array} \right]$$

The challenge





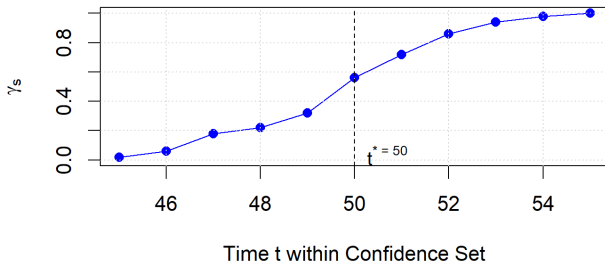
$$\hat{y}_{s,t} = \begin{cases} \hat{y}_{s,t}^j, & \text{if } c_{s,j-1}^{\max} \leq t \leq c_{s,j}^{\min}, \\ \gamma_{s,j,t} \hat{y}_{s,t}^j + (1 - \gamma_{s,j,t}) \hat{y}_{s,t}^{j+1}, & \text{if } c_{s,j}^{\min} \leq t \leq c_{s,j}^{\max}, \\ \hat{y}_{s,t}^{j+1}, & \text{if } c_{s,j}^{\max} \leq t \leq c_{s,j+1}^{\min}. \end{cases}$$





$$\gamma_{s,j,t} = \frac{\sum_{i \in A_t} \ell(i)}{\sum_{i \in \{c_{s,j}\}} \ell(i)},$$

where $A_t := \{y_{s,i} \in \{c_{s,j}\} : y_{s,i} \leq c_{s,j,t}\}$ and $\ell(i)$ is the log-likelihood of point i .



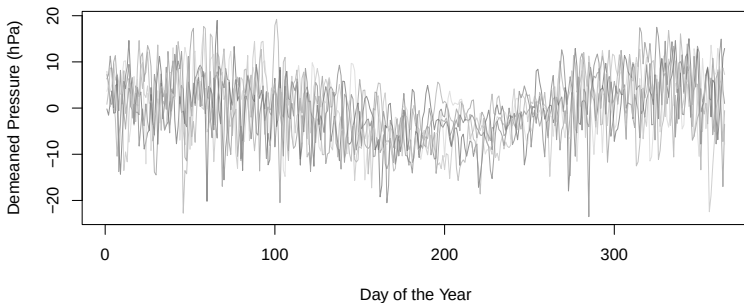


1. A method to detect changepoints
2. Confidence sets for those changepoints
3. A segment specific model to forecast from
4. A blending function (default is likelihood based)



Change in covariance

Demeaned Atmospheric Pressure (2018–2023)

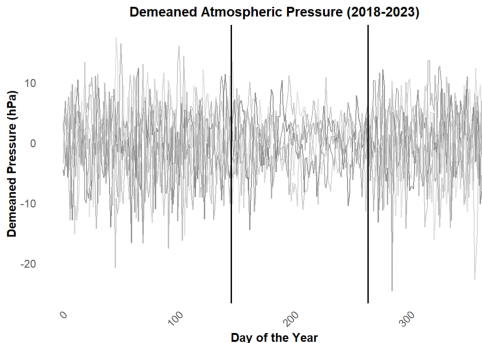


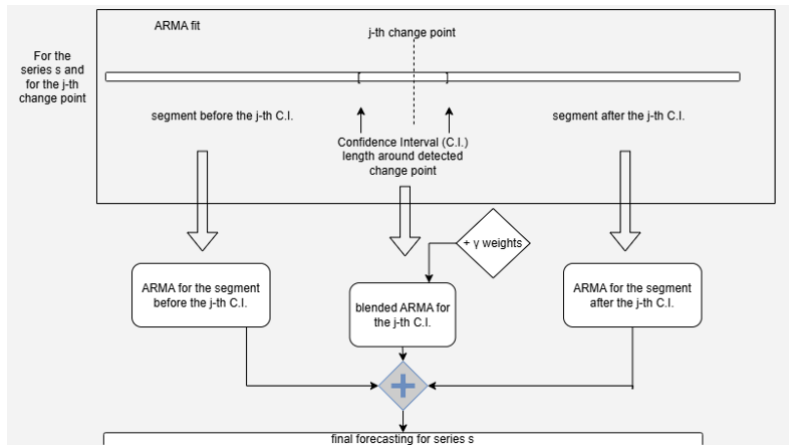
Example: AtmosPressure



Ryan, Killick (2023) Ratio: May 26 (146) [142,149], Sept 21 (264) [261,268]

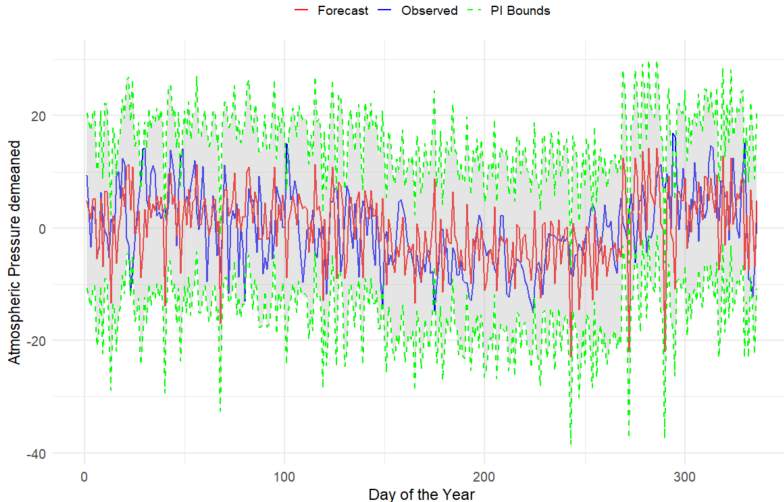
Li et al. (2023) BSCov: June 10 (161) [159,162], Aug 26 (238) [236,240]





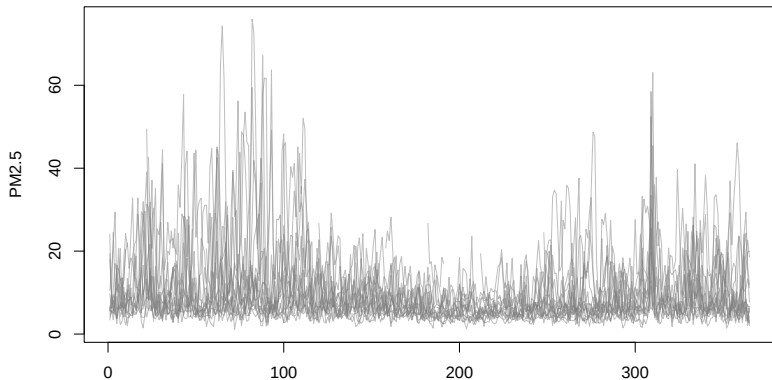


Observed vs Forecast with Prediction Intervals



Year	MSPE				MAPE(%)			
	Blend BSCov	Blend Ratio	Cpts	ARMA	Blend BSCov	Blend Ratio	Cpts	ARMA
2018	62.37	71.18	73.09	98.63	6.22	6.61	6.83	7.89
2019	65.79	61.91	68.67	77.01	6.50	6.26	6.63	7.06
2020	2.27e+15	66.37	78.21	x	1.33e+07	6.38	6.59	x
2021	95.17	73.83	72.98	92.94	7.96	6.79	6.69	7.85
2022	68.91	63.31	77.76	83.58	6.77	6.07	6.93	7.31

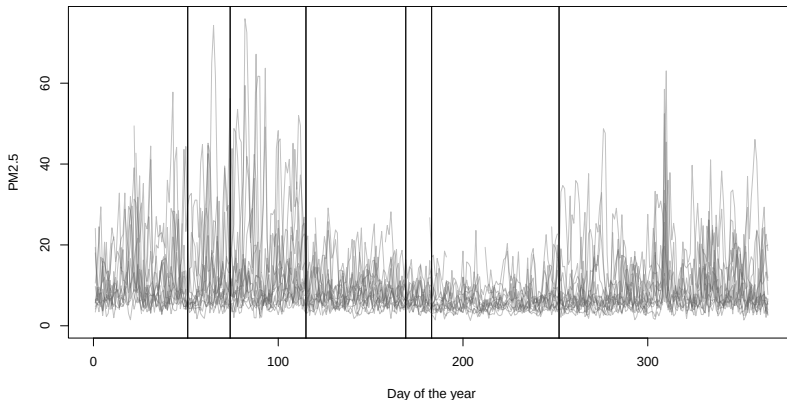
Example: PM2.5



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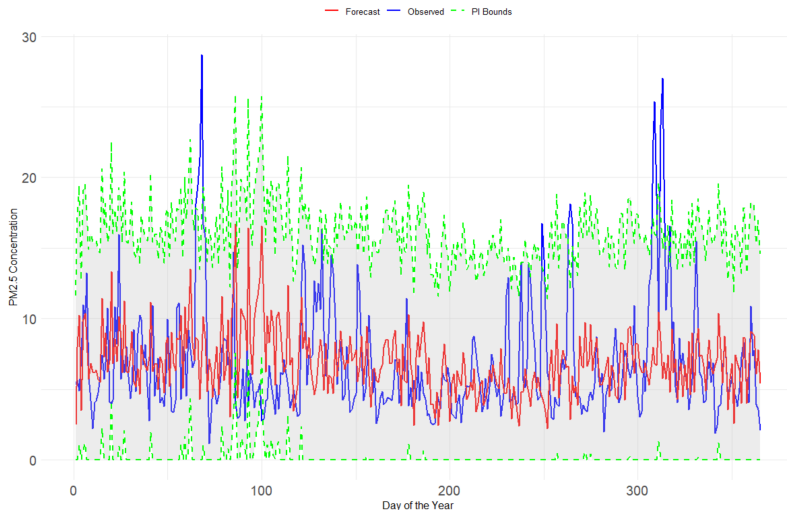
Univariate, change in rate



Cpt Date	Day of the year	95% C.S.
Feb 20	51	51
March 15	74	[65,76]
April 25	115	111, [114,122]
June 18	169	169
July 2	183	183, [188, 196], 198
Sept 9	252	[247,252]



Observed vs Forecast with Prediction Intervals



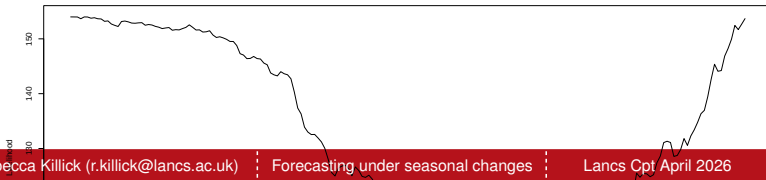
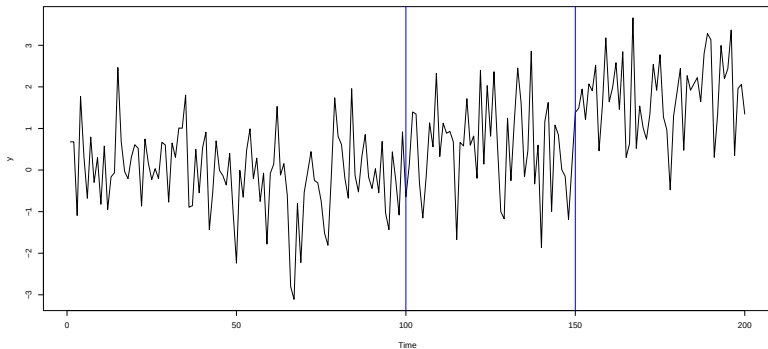
Horizon	MSPE				
	Cpts	ARMA	Blend AR(1)	Blend Struc	Blend MoSum
5-years	26.77	28.30	21.86	26.15	27.41
6-years	27.18	28.70	21.83	25.35	26.72

Horizon	MAPE (%)				
	Cpts	ARMA	Blend AR(1)	Blend Struc	Blend MoSum
5-years	3.57	3.80	3.05	3.42	3.22
6-years	3.60	3.84	3.05	3.37	3.17

Tools required:

1. A method to detect changepoints
2. Confidence sets for those changepoints
3. A segment specific model to forecast from
4. A blending function (default is likelihood based)

- A method for forecasting a new season of data
- Takes into account changepoints and uncertainties of changes
- More powerful than just considering the changepoints
- Agnostic to what model or changepoint approach you use





Two Changepoint Likelihood

